

Roll No...23T2028

I BE CLASS TEST-I MAR 2024
Electronics & Telecommunication
IAMRCI Applied Mathematics-I

Maximum Marks [20]

Time-70 mins.

Note-Attempt all the questions.

Q.1 Show that-

[4]

$$e^{\cos^{-1} x} = e^{\pi/2} \left(1 - x + \frac{x^2}{2} - \frac{x^3}{3} + \frac{5x^4}{24} - \frac{x^5}{6} + \dots \right)$$

Q.2 Find the radius of curvature at the point (r, θ) on the parabola-

[4]

$$2a/r = 1 + \cos \theta.$$

Q.3 Find the envelope of the family of parabolas-

[4]

$$y = x \tan \alpha - \frac{gx^2}{2u^2 \cos^2 \alpha},$$

α being a parameter.

Q.4 Find the angle between the surfaces $x \log z = y^2 - 1$ and $x^2 y = 2 - z$ at $(1, 1, 1)$.

[4]

Q.5 Find the divergence and curl of-

[4]

$$\vec{f} = (3x^2y - z)\hat{i} + (xz^3 + y^4)\hat{j} - 2x^3z^2\hat{k}.$$
